Curriculum vitae

Personal Information

FIRST NAME / SURNAME	Dr. Tatiana Morozova
DATE OF BIRTH	24.12.1992
NATIONALITY	Russian
CIVIL STATUS	married
EMAIL	tatiana.morozova@ens-lyon.fr

Education

DATES	Mar. 2016 - Oct. 2019
QUALIFICATION PROGRAM	**Ph.D. in Physics**. Thesis on “Polymers and Polymer Colloids in Solvent Mixtures”, under the supervision of Dr. A. Nikoubashman
INSTITUTION	Johannes Gutenberg University, Mainz, **Germany**
DATES	Sept. 2010 - Jan. 2016
QUALIFICATION AWARDED	**Specialist in Physics** (higher education program in Condensed Matter Physics equaled a Master's Degree). Thesis on “Computer Simulation of Mechanical Properties of Polyelectrolyte Micro- and Nanocapsules”, under the supervision of Prof. Dr. O. I. Vinogradova
INSTITUTION	Lomonosov Moscow State University, Moscow, **Russia**
DATES	Sept. 1999 - Jun. 2010
QUALIFICATION AWARDED	**High school Diploma**
INSTITUTION	Gymnasium № 1519, Moscow, Russia

Work Experience

DATES	Dec. 2024- *present*
POSITION	**Permanent** researcher (Chargée de Recherche)
INSTITUTION	CNRS, Physics Laboratory at ENS Lyon, Lyon, **France**
DATES	Dec. 2023 - Nov. 2024
POSITION	**Postdoctoral** researcher
INSTITUTION	CNRS, Institut Pasteur, Paris, **France**
DATES	Oct. 2020 - Nov. 2023
POSITION	**Postdoctoral** researcher
INSTITUTION	Institut Max von Laue - Paul Langevin (ILL), Grenoble, **France**
DATES	Nov. 2019 - Sept. 2020
POSITION	**Postdoctoral** researcher
INSTITUTION	LIPhy, CNRS, University Grenoble Alpes, Grenoble, **France**
DATES	Mar. 2016 - Oct. 2019
POSITION	**Ph.D.** student
INSTITUTION	Johannes Gutenberg University, Mainz, **Germany**

Teaching Experience

DATES March 2026
POSITION **Teaching assistant** for the course “Molecular dynamics” (M1)
INSTITUTION ENS Lyon, Lyon, France

DATES Winter semester 2017/2018 & Summer semester 2018
POSITION **Teaching assistant** for the course “Mathematics for Physicists”,
INSTITUTION Johannes Gutenberg University, Mainz, Germany

DATES Summer semester 2016
POSITION **Teaching assistant** for the course “Advanced Statistical Physics”,
INSTITUTION Johannes Gutenberg University, Mainz, Germany

Fundings Raised

DATES 2025-2029
NAME **Jeunes Chercheuses et Jeunes Chercheurs** (JCJC) starting grant for young scientists based in France (ANR): “In silico: Disordered Proteins in Cells and Bio-Inspired Materials” (€ 241,000).

DATES 2025
NAME **CNRS DIALOG**: computational equipment (€ 46,000)

DATES 2024
NAME **Marie Skłodowska-Curie Action** (MSCA) European Postdoctoral Fellowships, project: “In silico modeling of multi-domain proteins in biological condensates” (231,000€). *Declined* for the permanent research position at CNRS.

DATES 2022 - 2025
NAME Franco(**ANR**)-German(**DFG**) project: “Dynamics, kinetics and assembly of model intrinsically disordered proteins from a polymer physics perspective” in collaboration with F. Schreiber, M. Oettel, J.-L. Barrat, O. Matsarskaia, T. Seydel (650,000€).

DATES 2020 - 2026
NAME **Computational time** at the French national cluster facilities IDRIS (CNRS) (300,000 GPU hours granted ~ 110,000€)

DATES 2017
NAME **Travel grant** from the IRTG/TRR146 graduate school (3,000 €)

Research Visits

DATES July 2022
PURPOSE Collaborative visit to Dr. A. Deblais
INSTITUTION Institute of Physics, University of Amsterdam, **Netherlands**

DATES June - July 2017

PURPOSE Experimental project on fabrication of polymer colloids in the research groups of A.Z. Panagiotopoulos, R.K. Prud'homme, R.D. Priestley

INSTITUTION Department of Chemical and Biological Engineering, Princeton University, **USA**

Skills and Competences

LANGUAGES SPOKEN Russian (native), English (fluent), French (B1), German (intermediate)

PROGRAMMING C/C++, Python

SIMULATION SOFTWARE HOOMD-Blue, GROMACS, PyEmma, PLUMED, AlphaFold

Organization of Scientific Events

DATES Jul. 2021 - Oct. 2023

ROLE **College secretary** of the Theory Group - organization of seminars & group activities, research visits, and workshops at ILL

DATES Feb. - Oct. 2020

ROLE **Organizer** of group **seminars** of the Statistical Physics and Modeling team (LIPhy, UGA/CNRS)

DATES Oct. 2018 - Oct. 2019

ROLE **Ph.D. student representative** - organization of seminars

DATES May 2019

ROLE **Organization** of the graduate students (CRC TRR146) retreat (5 days, Amsterdam, Netherlands)

Scientific Service and Editorial Activities

DATES Sept 2026-present

ROLE **Guest Editor**, BMC Methods, Special Issue "*Computational biophysics and molecular dynamics simulations*"

ROLE **Reviewer** for *Biomacromolecules*, *PRE*, *ANR*

Publications

(1) **T.I. Morozova** and A. Nikoubashman: Coil-globule collapse of polystyrene chains in tetrahydrofuran-water mixtures, ***J. Phys. Chem. B*** 122, 2130 (**2018**)

(2) **T.I. Morozova**, V.E. Lee, A.Z. Panagiotopoulos, R.K. Prud'homme, R.D. Priestley, and A. Nikoubashman: On the stability of polymeric nanoparticles fabricated through rapid solvent mixing, ***Langmuir*** 35, 709 (**2019**)

(3) **T.I. Morozova** and A. Nikoubashman: Surface activity of soft polymer colloids, ***Langmuir*** 35, 16907 (**2019**)

(4) **T.I. Morozova**, V.E. Lee, N. Bizmark, S.S. Datta, R.K. Prud'homme, A. Nikoubashman, R.D. Priestley: In silico design enables the rapid production of surface-active colloidal amphiphiles, ***ACS Cent. Sci,*** 6, 166 (**2020**)

(5) **T.I. Morozova**, N.A. Garcia, J.-L. Barrat, G.S. Luengo, F. Leonforte: Adsorption and desorption of polymers on bio-inspired chemically structured substrates, ***ACS Appl. Mater. Interfaces***, 13, 25, 30086–30097 (**2021**)

(6) **T.I. Morozova**, N.A. Garcia, J.-L. Barrat: Temperature dependence of thermodynamic, dynamical, and dielectric properties of water models, ***J. Chem. Phys.***, 156 (12), 126101 (**2022**)

(7) M.R. Bittermann, **T.I. Morozova**, S.F. Velandia, A. Deblais, S. Woutersen, D. Bonn: Surface-mediated molecular transport in polydisperse oil-in-water emulsions, ***Langmuir***, 39, 12, 4207-4215 (**2023**)

(8) **T.I. Morozova**, N.A. Garcia, O. Matsarskaia, F. Roosen-Runge, J.-L. Barrat: Structural and Dynamical Properties of Elastin-Like Peptides near Lower Critical Solution Temperature, ***Biomacromolecules*** 24, 4,1912-1923 (**2023**); top twelve publications by researchers at the ILL (*Highlights* 2023)

(9) I. Adroher-Benítez, **T.I. Morozova**, N.A. Garcia, J.-L. Barrat, G.S. Luengo, F. Leonforte: Modeling the adsorption of copolymers on heterogeneous surfaces and the stability of polymer coatings for cosmetic applications, ***Macromolecules***, 56, 24, 10285-10295 (**2023**)

(10) **T.I. Morozova**, N.A. Garcia, J.-L. Barrat: Sequence length controls coil-to-globule transition in elastin-like polypeptides, ***J. Phys. Chem. Lett.*** ,15, 10757-10762 (**2024**)

(11) S. Chakraborty, **T.I. Morozova**, J.-L. Barrat: Intrinsically disordered proteins can behave as different polymers across their conformational ensemble, ***J. Phys. Chem. B,*** 129, 2359-2369 (**2025**)

(12) V. Schnapka, **T. I. Morozova**, S. Sen, and M. Bonomi: Atomic resolution ensembles of intrinsically disordered and multi-domain proteins with Alphafold, ***Nat. Commun.,*** 17, 2399 (**2026**)

(13) L. M. Miñarro, S. Chakraborty, C. Beck, A. C. Grundel, I. Mosca, F. Roosen-Runge , **T. I. Morozova**, J.-L. Barrat, F. Schreiber, and T. Seydel: Non-Fickian diffusion within assemblies of the intrinsically disordered protein β-casein", ***PNAS***, 123, 11, e2532636123 (**2026**)

(14) S. Sen, S. E. Hoff, **T. I. Morozova**, V. Schnapka, and M. Bonomi: Advancing in silico drug design with Bayesian refinement of AlphaFold models***, J. Chem. Theory Comput.,*** 22(14), 7407-7419 (**2026**)

Preprints

(*i*) F. B. Toplek, L. Borges-Araujo, K. Lindorff-Larsen, R. Everaers, P.

C. Souza, and **T.I. Morozova**: Towards transferable explicit-solvent coarse-grained models for biomolecular condensates, *BioRxiv* (**2026**)

Talks and Posters

Seminar: Computational Soft Matter Lab, University of Amsterdam, **2026**: *Disordered Proteins Across Scales: From Single-Chain Ensembles to Biomolecular Condensates*

Seminar: Laboratory of Theoretical Physics and Statistical Models, Orsay, **2026**: *Disordered Proteins: Insights from Polymer Physics*

Seminar: IPF-Dresden, **2024**: "*Thermoresponsive polypeptides: insights from the polymer physics*"

Seminar: Laboratoire de Physique ENS de Lyon, France, **2023**: "*Thermoresponsive polypeptides: insights from the polymer physics*"

Seminar: Laboratoire de biochimie théorique, IBPC, Paris, **2023**: "*Phase behavior of thermo-responsive polypeptides*"

Seminar: ICS Strasbourg, Strasbourg, **2022**, *Structural and Dynamical Properties of Elastin-Like Peptides near Lower Critical Solution Temperature*

Seminar: Thematics School in Soft Condensed Matter (Univ, Grenoble Alpes) **2021**, *Fabrication of Polymer Colloids through Rapid Solvent Exchange*

Seminar*:* University of Pennsylvania **2019***, Computer simulations of controlled fabrication of polymer colloids through the rapid solvent exchange*

Invited Talk: conference Coarse-Graining in the Age of Machine Learning and Molecular Design **2026** (Lyon, France): *Towards Transferable Explicit-Solvent Coarse-Grained Modelling of Biomolecular Condensates*

Invited Talk: conference FisMat **2025** (Venice, Italy): *Disordered proteins: insights from polymer physics*

Contributed Talk: CECAM workshop: Frontiers of Coarse-Grained Models: From New Developments to Modeling Dynamics, Assemblies, and Macromolecular Machines, Lyon (France), **2024**: *Thermo-responsive polypeptides: insights from polymer physics*

Contributed Talk: 28th International Conference on Statistical Physics (Statphys28), Tokyo (Japan), **2023**: *Coil-to-globule transition of thermoresponsive polypeptides*

Contributed Talk: Spring meeting of the American Chemical Society (online) **2021**, *Polymer adsorption on bio-inspired disordered substrate*

Contributed Talk: Spring meeting of the DPG, Regensburg (Germany) **2019**, *Fabrication of polymeric Janus nanoparticles and their behavior at the liquid-liquid interfaces - a simulation study*

Contributed Talk: Spring meeting of the DPG Berlin (Germany) **2018**, *Polymer self-assembly into nanoparticles through rapid solvent exchange in organic media*

Contributed Talk: 91st ACS Colloid&Surface Science Symposium, New York City (USA) **2017**, *Atomistic simulations of the coil-globule transition of polystyrene chains in solution*

Contributed Talk: Spring meeting of the DPG, Dresden (Germany) **2017**, *Controlled fabrication of nanoparticles through rapid solvent exchange*

Poster: Physics of Living Matter workshop, Cambridge (UK) **2026**: *Towards Transferable Explicit-Solvent Coarse-Grained Models for Biomolecular Condensates*

Poster: CECAM workshop: Development of Coarse-Grained Models, Lyon (France) **2022**, *Adsorption and desorption of polymers on bio-inspired chemically heterogeneous surfaces*

Poster: Les Houches-TSRC Protein Dynamics Workshop, Aussois (France) **2022**, *Structural and dynamical properties of elastin-like peptides near lower critical solution temperature*

Poster: CECAM workshop: From disordered biomolecular complexes to biological coacervates, Zurich (Switzerland) **2022**, *Structural and dynamical properties of elastin-like peptides near lower critical solution temperature*

Poster*: 5th* Edwards Symposium, Cambridge (UK/online) **2021**, *Adsorption and desorption of polymers on bio-inspired chemically heterogeneous surfaces* - **poster prize**

Poster: 3rd Edwards Symposium, Cambridge (UK) **2018,** *Fabrication of Janus nanoparticles from the self-assembly of homopolymers and amphiphiles through rapid precipitation*

Poster: CECAM workshop “Collective behavior of soft and active matter under confinement”, Mainz (Germany) **2018**, *Fabrication of Janus nanoparticles from the self-assembly of homopolymers and amphiphiles through rapid precipitation*